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Assessing Heat Stress Impacts on Cotton Yield and Revenue: A Multilevel Regression Approach in the Southern U.S.

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ABSTRACT

Climate change-induced heat stress significantly threatens cotton production in the Southern United States, reducing yields and farm revenue. This study quantifies the impact of rising temperatures using a multilevel regression model applied to historical climate and yield data (1980–2018) from key cotton-growing regions. The analysis examines how maximum and minimum temperatures, precipitation, and growing degree days influence yield at different growth stages. Results show that a 1°C increase in maximum temperature during the flowering stage (TmaxGS2) reduces cotton yield by 5.5%, leading to revenue losses of up to \$219 per acre. Conversely, higher precipitation during critical growth periods increases yield by 183 lb/acre, partially offsetting heat stress effects. Given these findings, adaptation strategies are essential. We recommend the development of heat-tolerant cotton varieties, improved irrigation management, and expanded financial support programmes, including climate-based crop insurance. Additionally, optimising planting schedules and adopting precision agriculture can help mitigate yield losses. These measures will strengthen the resilience of cotton farming against escalating climate risks, ensuring long-term sustainability. This study provides valuable insights for farmers, policymakers, and researchers working to safeguard cotton production in a changing climate.

1 | Background

Agricultural production is threatened by increased and excessive temperatures, especially in semi-arid regions where water resources are limited (Becker et al. 2023). The Intergovernmental Panel on Climate Change (IPCC) Fourth Assessment Report (2007) notes that a rise in greenhouse gas (GHG) emissions has resulted in an increase in average temperatures by 0.74°C since 1901, and this has caused a rise in global warming. The United States Environmental Protection Agency (2016) equally affirms that the most effective of these GHGs are carbon dioxide (CO₂), methane (CH₄), and nitrous

oxide (N₂O). However, it is pertinent to note that several studies have confirmed a potential negative effect of the impact of climate change on agricultural production, including cotton production (Ahmed et al. 2016). Cotton production, particularly in the Southern High Plains of Texas, is highly susceptible to weather risk due to a proclivity for drought and reduced rainfall. This situation, consequently poses a severe threat to crop yield (Mitchell et al. 2016; Wang et al. 2018).

Likewise, Reddy et al. (2000) aver that global climate change is expected to have a direct impact on agriculture, and the rate of change is estimated to be faster in 2100 than the previous

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1000 years. Given this, global climate change can result in heat stress, that is, excessive heat stress denotes the rise in temperature beyond a certain threshold level over a specific period, and which is enough to cause irremediable damage to plant growth and development (Wahid et al. 2007). Heat stress can also be described as high-temperature stress, which is notably one of the restraining factors in crop production (Majeed et al. 2021). However, temperature requirements vary from crop to crop depending on the intensity of exposure, air or soil temperature, and night/day temperature (Majeed et al. 2021). Thus, crops in different climatic regions have different degrees of temperature tolerance. Zahid et al. (2016) confirm instances of heat stress regarding the cotton grown in the United States and India, where temperatures above 38°C and 46°C, respectively, are recorded.

Although the heat stress impacts will vary in each state, but as one of the top agricultural exporters, there is a probability for the United States to experience an extremely high risk of heat stress by 2045 (Middleton and Will 2022). Relatedly, as the largest agricultural producer, and responsible for 21% of global food production in 2020, India is currently at a risk of heat stress owing to climate change (Middleton and Will 2022). Regarding India, the report published by the United States Department of Agriculture (USDA) in 2022 states categorically that heat stress resulting from an unprecedented spike in temperatures scorched the wheat harvested, which propelled the country to impose a ban on cotton exports. The ban evidently had a negative impact on the yield, leading to a decrease of about 10%–15%, while the market forecast for the year declined from 110 million metric tons (MMT) to 99 MMT. This was a blow to the international community, as India is the world's second-largest wheat producer, and expected to fill the supply gap caused by Russia's invasion of Ukraine. This resulted in a sudden spike in global wheat prices, in addition to posing a situation of food security risks to India.

Inherently, cotton under heat stress conditions faces several challenges from the planting stage to the harvesting stage. Consequently, each of these stages triggers a significant reduction in the yield of cotton fibre. Snider et al. (2009) affirm that the reproductive stages of cotton are more sensitive to high temperature than the vegetative growth stages. Similarly, as reported by Khan et al. (2020), high night temperature reduces cotton yield, causing significant damage, while the high day temperature is minor. Numerous studies have also testified to the impact of heat stress on cotton fibre yield (Abro et al. 2022; Reddy et al. 1999; Pettigrew 2008). Equally, He et al. (2013) discovered an increase of 2°C–3°C temperature from the optimal temperature (32°C) in Nanjing, China, which resulted in a 40% decrease in the yield of cotton. Apparently, heat stress significantly reduces cotton yield and yield-related parameters, including lint yield, boll weight, boll number, boll retention, seed yield, seed number per boll, and ginning percentage (Salman et al. 2019a).

Likewise, the research conducted by Sharma et al. (2022) on the impact of recent climate change on cotton and soybean yields in the southeastern United States adopting a fixed effect regression model, evidenced that 1°C of maximum temperature (heat stress parameters) decreased cotton yield by 10%, while 1°C of

minimum temperature decreased increase by 20%. Comparably, Reddy et al. (2002) projected that a 4°C increase in average growing season temperatures would reduce cotton yields by 9% in the Mississippi Delta. Salman et al. (2019b); Xu et al. (2020); and Yousaf et al. (2023) also predicted that heat stress will significantly reduce yield and yield-related parameters including lint yield, boll weight, boll number, boll retention, seed yield, seed number per boll, and ginning percentage in cotton. Thus, from the foregoing, it is apparent that the impact of heat stress is not limited to cotton yield but also on some other components of the plant.

Cotton is basically an important natural fibre that provides income for more than 250 million people across the globe and its production accounts for almost 7% of the labour in developing countries. In point of fact, approximately half of all textiles are made of cotton. Subsequently, the demand for cotton in the textile industry has grown steadily over the last decade because of increased population and income growth, especially in developing countries, particularly in Asia. Thus, the demand for cotton is a derived demand, as the global demand for textile products depends on its availability. Similarly, cotton is a major commodity crop in the United States, and the country is the third-largest cotton producer in the world, following China and India. Since the United States ranks third worldwide, it plays a key role in the global cotton market. Nevertheless, cotton production is threatened by heat stress in the country; hence, this current study is apropos. Intrinsically, temperature is an important factor that must not be sidelined because of its determinant role in assessing the rate of development in cotton (Bange and Milroy 2001). As aforesaid, excessive temperature is a threat because cotton crops respond to different levels of heat stress.

This has been studied across scales; nonetheless, when the level of heat stress is beyond a critical threshold (32°C), cotton yields are impacted negatively (Schlenker and Roberts 2009). Per se, the adverse effects of heat stress on cotton production have become a growing concern, both at the regional and global scale (IPCC 2014). As a result of climate change-induced warmer climates, the United States cotton production is projected to shift northward, a shift that is predicted to double the crop acreage (Elias et al. 2018). Thus, a systematic approach is needed to determine the impact of heat stress and the future of cotton production in the United States.

To this end, the overall objective of this study is to analyze the impact of heat stress on cotton yield in the Southern Region of the United States. Most of the studies related to heat stress in cotton have been carried out under greenhouses, growth chambers, or other controlled environmental conditions (Reddy et al. 1992; Brown and Oosterhuis 2004; Snider et al. 2009). However, to the best of our knowledge, no study has been conducted on the impact of heat stress on the various growth stages of cotton yield, particularly in the cotton-growing regions of the United States. Thus, a considerable gap exists regarding the comprehension of how cotton yield responds to heat stress across the southern United States. Therefore, this study will provide an insight into heat stress-induced responses on cotton yield, necessary for the identification of knowledge gaps and a plan for future lines of research, with a consequent development of heat-resilient cotton

for the United States and the world. This study is also expected to help the growers adapt to the changing climate, besides assisting in mitigating its adverse effects.

2 | Methods

This section expatiates the study area, data collection and consideration, estimation techniques, and empirical strategies.

Study area: The study was conducted across the primary cotton-growing regions of the United States, which were categorised into four regions encompassing 15 states. The categorisation was based on the geographical location within the southern United States, where cotton is predominantly produced. The regions include **Far West** (California, Arizona, and New Mexico), **Southwest** (Kansas, Oklahoma, and Texas), **Mid-south** (Missouri, Arkansas, Louisiana, Mississippi, and Tennessee), and **Southeast** (Alabama, Georgia, South Carolina, and North Carolina).

Data considerations: The dataset required for this study was collected from the cotton-growing regions of the United States. The crop yield data were obtained from The USDA. Historical data (from 1980 to 2018) containing the weather parameters from different cotton-growing regions of the United States were collected using GIS. The data was cleaned and subjected to statistical analysis using Stata software.

2.1 | Model Specification and Estimation Strategies

Multilevel Model: The impact of heat stress is determined by factors on different levels, ranging from the location or site level to the state level in the United States. A concerning problem is that there is a relationship between variables measured at different hierarchical levels, which poses a problem. Hox (1995) regards this as a multilevel problem. In this study, location or site-level factors and state-level determinants influence the yield of cotton; therefore, to handle this multilevel structure analysis, a single-level analysis like an ordinary regression model may be flawed Košmrlj and Širca (2014). Such a single-level model that prevails in the analysis of a linear model, in which the average cotton yield is continuous, is not appropriate. An analysis that is carried out at the location level may miss important group-level effects; a problem often referred to as the atomistic fallacy (Russo 2009). The classical linear regression model is considered inappropriate for analyzing unbalanced repeated measures and longitudinal data, as in this study. A simple panel data analysis with random or fixed effects is also not adequate because it does not allow for different slopes like multilevel models. Therefore, in this study, a two-level regression model was developed, based on the existing literature on cotton heat stress determinants. Factors describing a location's characteristics are included within the level (fixed effect), while factors affecting the states are incorporated between-level variables.

Similarly, according to Košmrlj and Širca (2014), a single-level analysis at the group level cannot be used to make assumptions at the individual (locations) level. Thus, locations or sites are

nested within states, and states are nested within regions, and they tend to share certain common values. Lastly, using a two-level model rather than a standard single-level regression has a big advantage in that it is possible to differentiate the two levels, providing fixed and random effects (Novak and Pahor 2017). With multilevel models, it is also possible to deal with complex variation at different levels. It does not treat locations or states as having the same error variance. Thus, this study explores a multilevel model to understand the heat stress determinants across the locations (sites) and states.

Estimation Strategies: Testing the evidence of substantial clustering on the dataset was the first step to ascertain the appropriateness of the mixed-level model. Notably, in a single level, we explore least squares regression with the main assumption that error terms are independent. So, the test on one of the indices to use is to determine the interclass correlation coefficients (ICC). This ICC was calculated by the ratio of upper-level variance estimates to the sum of lower-level and upper-level variance estimates. Conceptually, the ICC is like the R^2 effect size from the linear regression model. An ICC value > 0 indicates there is a clustering in the dataset. The more positive the value, the more evidence of substantial clustering existed in the dataset.

$$ICC = \frac{\text{Upper level variance estimates}}{\text{Lower level variance estimate} + \text{Upper level variance estimates}}, \quad (1)$$

$$p = \frac{\sigma^2(\gamma_{0j})}{\sigma^2(\mathcal{U}_{ij}) + \sigma^2(\gamma_{0j})}. \quad (2)$$

According to Abafe et al. (2021), the percentage of the variation in the score attributable to location characteristics is $1 - p$. If p is not too low, then it is necessary to add a first-level variable in the model, and if $1 - p$ is not very low it is necessary to incorporate first-level variables. Similarly, another way is to test the significance related to the variance of the upper level. This test is the likelihood ratio test compared to the linear model. If the test is found significant, this suggests that a mixed-level model is appropriate. In other words, estimating a blank model to see if locations or sites are different from states. The model can be expressed as follows:

$$\text{Location (Lower level) model : } Y_{ij} = \beta_{0j} + \mathcal{U}_{ij}, \quad (3)$$

$$\text{State (Upper level) model : } \beta_{0j} = Y_{00} + \gamma_{0j}. \quad (4)$$

The Intercept Y_{00} in this model above represents the fixed effect, which summarises the within-group regression, while γ_{0j} represents the random effect (between the groups) for the state level. A state-level variable to account for the variation was included in the model. Thus, estimating the following Random Intercept Model using upper-level predictors:

$$\text{Location (Lower level) model : } Y_{ij} = \beta_{0j} + \mathcal{U}_{ij}, \quad (5)$$

$$\text{State (Upper level) model : } \beta_{0j} = Y_{00} + Y_{01}W_j + \nu_{0j}, \quad (6)$$

$$\text{Full model : } Y_{ij} = Y_{00} + Y_{01}W_j + \nu_{0j} + u_{ij}. \quad (7)$$

Where:

W denotes the state-level variable.

Y_{00} and Y_{01} are the fixed effects,

and ν_{0j} is the random effect.

Then, the location parameters or attributes X were included in the model to account for their effect in the variation of evaluation. The matching model is a Random Intercept Model with both lower and upper level regressors. It can be expressed as follows:

$$\text{Location Lower level model : } Y_{ij} = \beta_{0j} + \beta_{1j} X_{ij} + u_{ij}, \quad (8)$$

$$\text{Full model : } Y_{ij} = Y_{00} + Y_{01}W_j + Y_{10}X_{ij} + \nu_{0j} + u_{ij}. \quad (9)$$

Therefore, Full Maximum Likelihood (FML) estimation was employed to fit the model. The mixed-level model was specified as follows:

$$\begin{aligned} \ln \text{Average yield}_{it} = & B_0 + B_1 \cdot \ln \text{Prct}_{it} + B_2 \cdot T \max \text{GS1}_{it} \\ & + B_3 \cdot T \max \text{GS2}_{it} + B_4 \cdot T \min \text{GS1}_{it} \\ & + B_5 \cdot T \min \text{GS2}_{it} + B_6 \cdot T \min \text{GS3}_{it} \\ & + \sum_j B_j \cdot \text{year}_i + u_i + \epsilon_{it}, \end{aligned} \quad (10)$$

where:

B_0 is the intercept term

$B_0, B_1, \dots, B_6$ are the coefficients of the fixed effects for the main climatic variables.

$\sum_j B_j \cdot \text{year}_i$ represents the sum of the coefficients for the year dummy variables (e.g., 2005, 2006, ...2018), with each year j capturing the unique effect for that year. The base year (2010) is omitted as the reference category.

u_i captures the random effects for each state (i), accounting for unobserved heterogeneity.

ϵ_{it} is the error term for each state-year observation.

Table 1 shows that employing a multilevel regression model is suitable. The intraclass correlation coefficient (ICC) for *State_Id* is 0.1219, indicating that approximately 12.2% of the total variation in the dependent variable is attributable to differences between states. The standard error of the ICC is 0.0568, suggesting a modest degree of clustering at the state level and justifying the use of multilevel modelling in the analysis.

TABLE 1 | Intraclass correlation.

Level	ICC	Standard error
State_Id	0.1219	0.0568

Cotton plants have an indeterminate growth habit, making their sensitivity to adverse environmental conditions exceptional. Their growth and development follow a typical sigmoid curve (NCC 1996). Thus, the study categorised cotton growth and fibre development into distinct and overlapping developmental stages. Following Ritchie et al. (2007), as illustrated in Figure 1 below, the phenological stages of cotton growth are divided into three main growth stages: Growth Stage 1 (GS1), Growth Stage 2 (GS2), and Growth Stage 3 (GS3):

- **GS1:** Germination to early vegetative growth (Days 1–60).
- **GS2:** Flowering begins, and cotton bolls start to develop (Days 61–100).
- **GS3:** Cotton bolls begin to open and reach full maturity (Days 101–160).

Figure 1 shows the progression of cotton development across three key growth stages: GS1 (Days 1–60), representing germination to early vegetative growth; GS2 (Days 61–100), when flowering and boll formation begin; and GS3 (Days 101–160), when bolls open and the crop reaches maturity. These stages are critical windows for assessing environmental impacts, such as heat or moisture stress on cotton yield and development.

Thirdly, the Variance Inflation Factor (VIF) test and mean decentralisation were performed to address multicollinearity issues. Natural logarithms were applied to account for outliers and resolve heteroskedasticity problems in the model. Additionally, a quadratic term (the square of Tmax and Tmin) was included to capture the non-linear relationships between these temperature variables. Similarly, a set of dummy variables was introduced to represent different years (13 dummies for 2005–2018). The year 2010 was used as the base category because it had the highest average cotton yield compared to other years in the study period.

Economic impact estimations: To assess the economic impact of heat stress on cotton yield, we began by calculating total revenue through the multiplication of cotton yield by prevailing market prices. This initial revenue computation provided a baseline for understanding the financial implications of yield variations. Subsequently, we employed a predictive model that integrated unit changes in our explanatory variables to analyze yield changes over time. By systematically examining these yield changes alongside market prices, we generated a new revenue estimate that reflects the influence of climatic conditions on overall revenue. By comparing the new revenue estimates with the initial calculations, we were able to quantify the economic impact of heat stress on cotton yield more accurately. This approach highlights the critical relationship between climate-induced yield fluctuations and revenue, providing valuable insights for stakeholders in the agricultural sector and policymakers aiming to mitigate the adverse effects of climate change on crop production. It is worth noting that our estimate was achieved by fitting the model and predicting how much $\hat{y}$ (cotton yield) changes when a key variable (e.g., TmaxGS2) changes by 1 unit while holding other variables constant. This gives the expected change in the average cotton yield based on the estimated coefficients in our initial model. This prediction is a point estimate and does not account for any uncertainty or

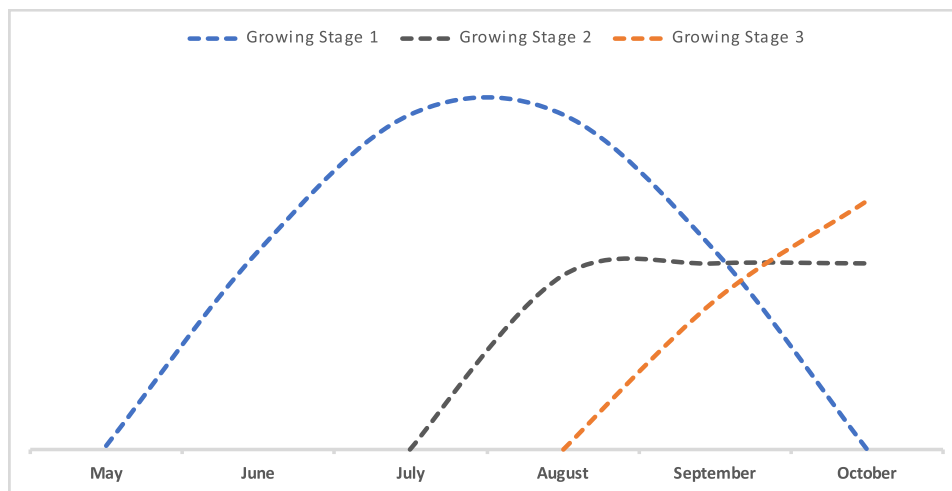


FIGURE 1 | Phenological stages of cotton growth.

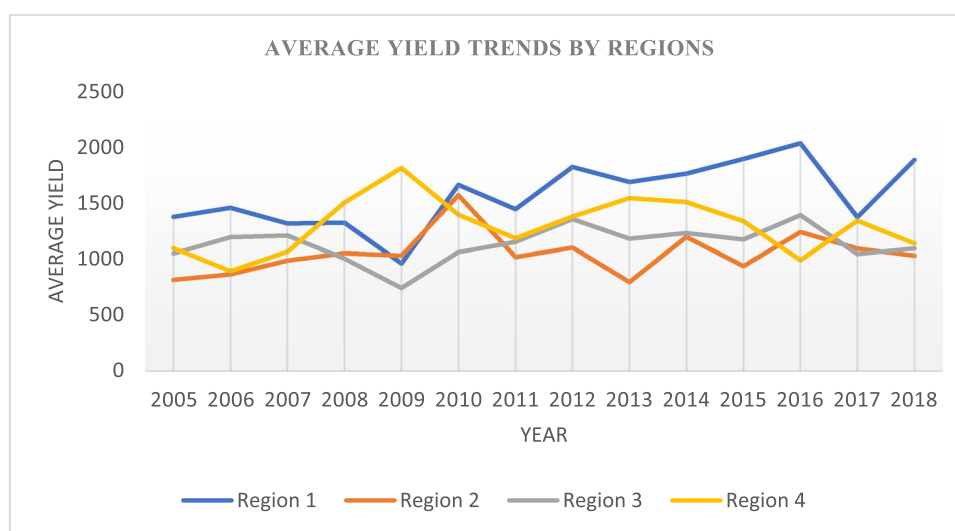


FIGURE 2 | The trends for the average yield in the regions.

variability around the prediction. To consider the uncertainty associated with the estimated $\hat{y}$ (cotton yield), we incorporated the standard error into predictions. The standard error is associated with the predicted values and the regression coefficients and gives a measure of how much the predictions might vary due to sampling variation. In other words, instead of providing a single point estimate, we calculated a prediction interval (which is broader than a confidence interval) that gives a range in which the actual value of the dependent variable is likely to fall, given the variability in our model. This is particularly useful for assessing the reliability of our predictions.

3 | Result and Discussion

3.1 | Descriptive Statistics

Supporting Information S1: Table S2 shows the descriptive statistics where the average cotton yield in the Southeastern United States (SE-US) is 1177.96, and the average minimum temperature (Tmin) and average maximum Temperature

TABLE 2 | Hotelling's *T*-squared test.

Hotelling T2	Hotelling F (3, 35)	Prob > F
49.65	15.66	0.0000

(Tmax) are 18.04°C and 31.15°C, respectively. The average growing degree days or degree units (GDD), which measure heat accumulation, are 9.58°C.

As indicated, Figure 2 above and Supporting Information S1: Table S1 show the average yield comparison across the regions, while Table 2 articulates the Hotelling's *T*-squared test (which was conducted to evaluate whether the mean vectors of the variables differ significantly across regions). The null hypothesis posits that all mean vectors are equal. Therefore, the results yielded a Hotelling's *T*-squared statistic of 49.65 and a corresponding Hotelling *F*-statistic of 15.66 with degrees of freedom (3, 35). The associated probability value (Prob > *F*) is extremely low (0.0000), providing strong evidence against the null

hypothesis and indicating significant differences in mean vectors across the regions. This suggests that at least one region significantly differs from the others in terms of the combination of dependent variables under consideration.

Estimate results from the multi-level regression.

The estimation from exploring the multi-level regression model with random intercepts using fixed effects and the random effects.

Fixed effects:

The fixed effects portion of the model refers to the coefficients that are assumed to be constant across all levels of the hierarchy, that is, across all the regions. This part explains the regression coefficients that estimate the relationship between the predictor variables and the outcome variable, assuming these relationships are the same for all groups or clusters in the data. The study presented two tables that show the effect or the relationship between the average yield of cotton on Tmax and Tmin, as well as GDD (Growing degree days). This was calculated from both the Tmin and Tmax. Supporting Information S1: Table S3 enunciates the summary of the regression analysis, where significant variables that affect the average yield are reported. It should be noted that multicollinearity was considered during the study, and such variables were removed from the model.

The analysis showed that rainfall (precipitation) affected average cotton yield marginally and negatively significantly. This suggests that a 1% increase in the unit of rainfall (mm/day) at the mean decreases cotton average yield by approximately 10% ($p < 0.1$). The result evinces that rainfall is less needed, especially during the late or last growing stage. This is in conformity with Parvin (2005), as the research posited that a unit centimetre accumulation of rainfall during the late season resulted in a greater yield reduction of 2.09 kg of lint than the same amount of rainfall during the early season. Similarly, Chen et al. (2017) affirmed that a decrease in average cotton production due to the change in precipitation was approximately 1.1%.

It is important to note that specific temperature thresholds for heat stress can vary based on the cotton variety, local climate conditions, and the crop's growth stage. Maximum temperature (Tmax) in the GS2 was found to be negatively significant at 10% ($p < 0.1$). This indicates that at the mean, a 1°C increase in average maximum temperature in the GS2 decreases the average cotton yield by approximately 5.5%. Given this, the result buttresses the crucial role of temperature in various physiological processes, viz., a central determinant factor in plant growth and development. In this study, the average Maximum temperature in GS2 (TmaxGS2) as portrayed in Supporting Information S1: Table S2 is 34.20°C. Additionally, the duration of exposure to high temperatures is a critical factor in determining the extent of heat stress impact. Similarly, Schlenker and Roberts (2009) confirmed that temperature plays a vital role in the growth of cotton, with a critical threshold of 32°C beyond which yields are adversely impacted. This study is also in line with Chen et al. (2017), that changes in temperature and precipitation decreased cotton yields in China. Furthermore, Reddy et al. (1999) and Snider and Oosterhuis (2012) attested to the

fact that cotton is less sensitive to heat stress at the vegetative stage, while flowering and boll development are the most sensitive stages.

Consequently, as depicted in Supporting Information S1: Table S4, the growing degree days in the ever-increasing stage 1 (GS1) were positive and statistically significant at 5% ($p < 0.05$). This result signifies that, at the mean, a 1°C increase in growing degree days or heat units in the GS1 increases the average cotton yield by approximately 7.37%. Similarly, the growing degree days in the GS2 decrease the average cotton yield. At the mean, a 1°C increase in growing degree days in the GS2 decreases the average cotton yield by approximately 5.92%. The result was negative and statistically significant at 1% ($p < 0.1$). The amount of heat absorbed by the cotton plant in the GS1 positively affects or increases the average cotton yield. In contrast, the average cotton yield in the GS2 decreases as the amount of heat absorbed by the cotton plant increases. This signifies that, in growing stage 1, where the vegetative process of the plant occurs, it needs more heat energy and is less sensitive to heat stress, compared to GS2. This discovery paralleled Wright et al. (2005) and Sharma et al. (2022), that accumulated growing degree days required for the cotton crop to reach different growth stages varies and affects average cotton yield at various growth stages.

In the dataset, 2010 represents the variable year; it is a dummy or categorical variable (each year is a dummy). Comparing the relative-based category (2010) against 2005–2018 depicted that the highest cotton yield was realised in 2010. Consequently, the year serves as a baseline for understanding the effects of temperature changes over time. By utilising 2010 as the base category, it implies that this year is a representative point of comparison for the evaluation of the impacts of heat stress in subsequent years. Besides the fact that yields were high in 2010, indicating favourable growing conditions for cotton, the contrast between this year and others allows us to capture the long-term trends of heat stress and its economic effects. As shown in Supporting Information S1: Table S3, we found out that 2005, 2006, 2007, 2011, and 2013 were statistically significant and distinctly different from 2010.

The distinction could be attributed to the fact that relative to the base category (the year 2010), the average cotton yield decreased in 2005, 2006, 2007, 2011, and 2013 by 33.17% ($p < 0.05$), 35.290% ($p < 0.01$), 25.95% ($p < 0.1$), 30.11% ($p < 0.05$) and 30.25% ($p < 0.05$), respectively. In contrast to the analysis displayed in Supporting Information S1: Table S4, we discovered that 2005, 2006, 2011, and 2013 were statistically significant and different from 2010. Relative to the base category (the year 2005), the average cotton yield decreased in 2005, 2006, 2011, and 2013 by 30.97% ($p < 0.05$), 37.21% ($p < 0.01$), 30.11% ($p < 0.05$) and 28.95% ($p < 0.05$), respectively. This could probably be attributed to the fact that the yield in 2010 was higher compared to other years. Over the years, especially the significant years, there have been improvements in cotton seed varieties and seed breeding. Many varieties and technological advances towards cotton production for better yield have been explored. Research targeting irrigation optimisation and strategies to combat heat stress has been prevalent. Similarly, different improved management practices also contribute to the increasing cotton yield amid heat stress and weather variability in those years.

3.2 | Random Effects

The random effects part of the model shows the variance of the intercepts and the residuals. As represented in Supplementary Table 3. The estimated variance of the random intercept is 0.0260. This indicates the amount of variability in the intercepts among the groups after accounting for fixed effects. In other words, it suggests how much the intercepts differ from the overall mean intercept. A higher variance would indicate more variability in the intercepts across groups. There is a moderate amount of variability in the intercepts among the groups, and the model is quite appropriate and consequently accurate in this estimate. The estimated variance of the residual from the random intercept is 0.2057. This represents the variance of the residuals (within-group variance). However, the model could not account for the variation in the outcome variable. Per se, this indicates the amount of variability in the individual observations that is not accounted for by the fixed effects or the random intercept. Basically, it reflects the unexplained variability at the level of individual observations. Similarly, in Supporting Information S1: Table S4, the estimated variance of the random intercept is 0.0320 while the estimated variance of the residual from the random intercept is 0.2039, following the same interpretation as above.

3.3 | Heat Stress Impact and Average Yield Comparison Across the Regions

The study further disaggregates the entire dataset consisting of the Southeastern United States (SE-US) cotton-producing area into regions as shown in Figure 2 above. Several pool regression models with state-fixed effects and location-fixed effects were explored, in which a regression with state-fixed effects outperformed the other results. Thus, to compare the effect of heat stress on the average cotton yield across the regions, the pool regression fitting a quadratic regression model with state-fixed effects was explored. Pertinently, the model accommodates the quadratic nature of the explanatory variables to capture the non-linear relationships which follow a parabolic (U-shaped or inverted U-shaped) pattern. Similarly, the inclusion of the quadratic terms provides more flexibility in the model, as it could represent more complex patterns of association between variables. Consequently, the inclusion facilitates a good comprehension and a subsequent prediction of the behaviour of the dependent variable. Lastly, the state-fixed effect in the model was used to compare the average cotton yield across the states with the base category state in each region. A summary of the estimates and comparisons across the regions is articulated in Supporting Information S1: Table S5. Each region is taken in turn.

Region 1:

In Region 1, the result shows that the Maximum temperature (TmaxGS3) in the growing stage 3, Solar radiation and Vapour Pressure were statistically significant at 10%, 1%, and 10%, respectively.

The maximum temperature (TmaxGS3): The investigation depicts that the coefficient for the linear term (TmaxGS3) is 0.0879. This implies that, at the mean for each unit (degree C)

increase in TmaxGS3, the average cotton yield increases by 87.9 lb/acre, consequently holding the quadratic term constant. Also, the coefficient for the quadratic term (Supporting Information S1: Table S5 shows the regional average yield trend by states) (TmaxGS3²) is −0.0124. This negative coefficient signifies a concave (inverted U-shaped) relationship between TmaxGS3 and the average cotton yield. Specifically, as TmaxGS3 increases, its effect on average cotton yield increases up to a certain point and then starts to decrease. However, that it is non-significant implies that the quadratic impact might not be strong enough to conclude a non-linear relationship definitively. By implication, the model denotes an initial positive effect of TmaxGS3 on the average cotton yield up to a certain point, after which the effect becomes negative, forming an inverted U-shaped relationship. This type of model is useful in identifying optimal conditions or thresholds, beyond which additional increases in TmaxGS3 become detrimental.

The maximum temperature (TmaxGS1): The coefficient for the linear term (TmaxGS1) is −0.0170. This implies that, at the mean for each 1°C increase in TmaxGS1, the average cotton yield increases by 17 lb/acre, holding the quadratic term constant. However, the linear term was not statistically significant, indicating that the evidence that TmaxGS1 has a linear effect on the average cotton yield is not strong enough to rule out random chance. However, the quadratic term was found statistically significant, in which the coefficient for the quadratic term (TmaxGS1²) is −0.0174. This negative coefficient suggests a concave (inverted U-shaped) relationship between TmaxGS1 and the average cotton yield. Intrinsically, as TmaxGS1 increases, its negative effect on the average cotton yield becomes more pronounced. By implication, the model indicates an inverted U-shaped relationship where average cotton yield initially decreases with increasing TmaxGS1, reaching a minimum value of around 0.49 units of TmaxGS1, after which average cotton yield starts to decrease more rapidly. This highlights the importance of considering both linear and quadratic terms to capture the true nature of the relationship between TmaxGS1 and the average cotton yield.

Solar radiation: The study suggests that the coefficient for the linear term (Solar radiation) is −0.0331. This means that, on average, for each one-unit increase (watts per square metre [W/m²]) in Solar radiation, the average cotton yield decreases by 33.1 lb/acre, holding the quadratic term constant. In the same vein, the coefficient for the quadratic term (Solar radiation²) is 0.0024. This positive coefficient suggests a convex (U-shaped) relationship between Solar radiation and the average cotton yield. Naturally, as Solar radiation increases, its negative effect on average cotton yield lessens until it gets to a certain point where the relationship turns positive. By implication, the model indicates an initial decrease in average cotton yield with increasing Solar radiation up to a certain point, after which average cotton yield starts to increase. This U-shaped relationship is a signification that there is an optimal range of Solar radiation beyond which its effect on average cotton yield changes from negative to positive.

Vapour pressure: The result indicates that, at the mean, the coefficient for the linear term for each one-unit increase in Vapour Pressure, the average cotton yield decreases by

1.2 lb/acre in turn. Considering the quadratic term, the small negative coefficient signifies a slightly concave (inverted U-shaped) relationship between Vapour Pressure and the average cotton yield. Demonstrably, as Vapour Pressure increases, its negative effect on average cotton yield becomes slightly more pronounced. By implication, the model indicates an initial slight decrease in the average cotton yield with increasing Vapour Pressure up to a certain point (1894.13 units), after which the average cotton yield starts to decrease more rapidly. This inverted U-shaped relationship highlights the importance of considering both linear and quadratic terms to capture the true nature of the relationship between Vapour Pressure and the average cotton yield.

Region 2:

In Region 2, the variable year, and the average temperature in growth stages 1, 2, and 3 are all statistically significant at 1%, 1%, 1%, and 10%, respectively.

The maximum temperature (TmaxGS1): The result from the study shows a significant positive linear effect, indicating that increases in TmaxGS1 are associated with increases in the average cotton yield. Although the quadratic term is not significant, its positive sign presupposes a trend towards an accelerating increase in the average cotton yield as TmaxGS1 increases. Since the quadratic term is positive, the relationship between TmaxGS1 and the average cotton yield forms a parabola that opens upwards. As TmaxGS1 increases, the average cotton yield increases at an accelerating rate due to the positive quadratic term.

The maximum temperature (TmaxGS2): The average cotton yield during GS2 increases as the maximum temperature rises. The coefficient for the linear term (TmaxGS2) suggests that, on average, each 1°C increase in TmaxGS2 leads to a decrease of 79.4 lb/acre in cotton yield, assuming the quadratic term remains constant. The quadratic term in this model is not statistically significant, indicating that there is insufficient evidence to suggest a quadratic effect of TmaxGS2² on average cotton yield beyond random variation. Overall, the results indicate a negative linear relationship between TmaxGS2 and average cotton yield, with a potential (though statistically weak) U-shaped pattern, where yield decreases to a minimum of around 4.34 units of TmaxGS2 before increasing again.

The maximum temperature (TmaxGS3): The maximum temperature in the growing stage 3 (TmaxGS3) increases with the coefficient of the linear term (TmaxGS3), which is 0.0604. This indicates that, at the mean, for each 1°C increase in TmaxGS3, the average cotton yield increases by 60.4 lb/acre, assuming the quadratic term is constant. The interpretation of the quadratic term (TmaxGS3²) is 0.0206 and positive. Congruently, the coefficient suggests a convex (U-shaped) relationship between TmaxGS3 and the average cotton yield. As TmaxGS3 increases, the average cotton yield initially increases at a decreasing rate, reaches a turning point, and then starts to increase at an increasing rate due to the positive quadratic term.

Year: The positive and significant coefficient implies that for each additional year, the average cotton yield increases by

approximately 30.6 lb/acre. This result indicates a positive progression over time, providing important insights into the temporal dynamics of the studied phenomenon.

Region 3:

Supporting Information S1: Table S5 shows a pooled regression obtained from Region 3. TmaxGS1, TmaxGS3, Precipitation, Solar radiation, and Tmin were statistically significant at 10%, 5%, 1%, 1%, and 1%, respectively.

The maximum temperature (TmaxGS1): With respect to the coefficient of the linear term, the analysis denotes that for each 1°C increase in TmaxGS1 at the mean, the average cotton yield increases by 73.6 lb/acre, assuming the quadratic term is constant. The coefficient of the quadratic term (TmaxGS1²) is negative (−0.0715), consequently inferring a concave (inverted U-shaped) relationship between TmaxGS1 and average cotton yield. Precisely, the model suggests an initial positive relationship between TmaxGS1 and average cotton yield, with a peak of around 0.514 units of (Supporting Information S1: Table S5 shows the Regional Weather Parameters across the regions) TmaxGS1, after which the relationship becomes negative. Subsequently, this underpins the significance of considering both linear and quadratic terms to fully understand the nature of the relationship between TmaxGS1 and the dependent variable.

The maximum temperature (TmaxGS3): The coefficient for the linear term (TmaxGS3) is 0.1017. This indicates that, at the mean, for each 1°C increase in TmaxGS3, the average cotton yield approximately increases by 101.7 lb/acre, assuming the quadratic term is constant. Although the quadratic term is not significant, its positive sign suggests a trend towards an accelerating increase in the average cotton yield as TmaxGS3 increases.

Precipitation: The linear term of precipitation is statistically significant, indicating a strong negative linear relationship with the average cotton yield. This result indicates that, for each one-unit increase in Precipitation at the mean, the average cotton yield decreases by 148.7 lb/acre, assuming the quadratic term is constant. The quadratic term is not statistically significant, suggesting weaker evidence for a quadratic effect. Since both the linear and quadratic terms are negative, the relationship between precipitation and average cotton yield forms a parabola that opens downwards. As Precipitation increases, the average cotton yield initially decreases at a linear rate, and the rate of decrease might accelerate due to the negative quadratic term.

Solar radiation: The coefficient for the linear term (Solar radiation) at the mean indicates that for each one-unit increase in Solar radiation, the average cotton yield decreases by 22.9 lb/acre, provided the quadratic term is constant. The combined effect indicates that both the linear and quadratic terms are negative; the relationship between Solar radiation and average cotton yield forms a parabola that opens downwards. As Solar radiation increases, the average cotton yield decreases at a linear rate, and the rate of decrease accelerates due to the negative quadratic term.

The minimum temperature (Tmin): Tmin represents the average minimum temperature across all the growing stages. The

coefficient for the linear term (T_{min}) is -0.2669 . This is an indication that, at the mean, for each one-unit increase in T_{min} , the average cotton yield decreases by 266.9 lb/acre, assuming the quadratic term is constant. The coefficient for the quadratic term (T_{min}^2) is -0.0245 . This negative coefficient suggests a concave (inverted U-shaped) relationship between T_{min} and the average cotton yield. In essence, as T_{min} increases, the negative effect on the average cotton yield becomes more pronounced. Inferentially, the model infers a decreasing relationship between T_{min} and the average cotton yield, with a linear decrease in the average cotton yield as T_{min} increases. The quadratic term, though not significant, indicates a potential accelerating decrease in the average cotton yield with higher levels of T_{min} .

Region 4:

In Region 4, the maximum temperature in the GS2 and precipitation were statistically significant at 10% ($p < 0.1$).

The Maximum temperature (T_{maxGS2}): the coefficients for T_{maxGS2} and T_{maxGS2}^2 in a regression model, the coefficient for the linear term (T_{maxGS2}) signifies that, at the mean, for each 1°C increase in T_{maxGS2} , the average cotton yield decreases by 90.9 lb/acre, on the assumption that the quadratic term is constant. The quadratic term is not significant; its negative sign indicates a trend towards an accelerating decrease in average cotton yield as T_{maxGS2} increases. Based on that, the model implies a decreasing relationship between T_{maxGS2} and the average cotton yield, with a linear decrease in the average cotton yield as T_{maxGS2} increases.

Precipitation: The linear term of precipitation is not statistically significant, indicating that the model has a non-linear relationship between Precipitation and the average cotton yield. On the other hand, the quadratic term was statistically significant at 10%, which indicates that the relationship between Precipitation and average cotton yield forms a parabola that opens downwards. As Precipitation increases, the average cotton yield initially increases at a linear rate, but after a certain point, the positive effect diminishes and turns negative due to the quadratic term.

State fixed effect:

The study estimates the state-fixed effect across the regions as shown in Supporting Information S1: Table S5, except for Region 1 where multicollinearity exists. In Region 2, Texas State was compared to Oklahoma State as a category. Following this, the result of this region suggests that relative to Oklahoma State, the average cotton yield in Texas decreased by approximately 658.6 lb/acre, and it was statistically significant at 1% ($p < 0.001$). Similarly, in Region 3, relative to Arkansas State, Louisiana State was statistically significant at 5% ($p < 0.05$), and the average cotton yield was approximately increased by 331.7 lb/acre. Likewise, the average cotton yield in Mississippi was 443.4 lb/acre, higher than that in Arkansas State, and the result was statistically significant at 1% ($p < 0.001$). Also, Tennessee's average cotton yield was higher than Arkansas's by approximately 243.8 lb/acre and was statistically significant at 10%. In Region 4, relative to Alabama, a base category as shown

in Supporting Information S1: Table S5, the average yield of cotton in Georgia State increased by approximately 432.8 lb/acre and was statistically significant at 1%. Similarly, the average cotton yield in Virginia State increased by 714.7 lb/acre at 1% ($p < 0.001$).

3.4 | Economic Implications of the Heat Stress Impact on the Average Cotton Yield and Revenue

This section shows the result of the economic prediction of cotton yield with a 1°C increase in temperature and mm/day. In Table 4, the analysis of cotton yield and revenue under varying temperature and precipitation conditions across multiple years reveals significant economic implications. Maximum temperature increases during the growing stages (T_{maxGS2}), demonstrating a clear relationship between reductions in both yield and revenue.

T_{maxGS2} Impact: The results show consistently negative yield and revenue changes for T_{maxGS2} , pointing to severe yield losses due to prolonged heat stress later in the growing stage. For instance, in 2010, cotton yield dropped by 268 lb/acre, and revenue by \$219/acre. The significant decrease in both emphasises the detrimental effect of sustained high temperatures. This trend holds across most years, including 2018. The year exhibited similar reductions in both yield and revenue, cotton yield manifested a decline of 250 lb/acre and a significant reduction in revenue: \$176/acre.

Precipitation (Prcp) Impact: Precipitation also plays a critical role in mitigating the effects of heat stress. Higher precipitation levels in 2016 resulted in a yield increase of 183 lb/acre and revenue gains of \$123/acre, demonstrating how moisture availability can cushion the impact of high temperatures. Conversely, years with lower precipitation, such as 2007, show significantly lower yield improvements (3 lb/acre) and minimal revenue benefits (\$1.80/acre), suggesting that water stress exacerbates the economic losses from heat stress. This is captured in Table 3, and it presents a comprehensive synopsis of changes in cotton yield and revenue between 2005 and 2018 with respect to the germaneness of precipitation in lessening the contrary impact of heat stress.

Table 4 below summarises the statistics for yield and revenue changes in response to variations in maximum temperature during the second growth stage (T_{maxGS2}) and precipitation (Prcp). The negative mean value for $\text{Yield_change_Tmax2}$ (-183.04 lb/acre) implies that increased T_{maxGS2} is associated with a decline in cotton yield. Equally, the standard deviation (44.02) indicates variability in this decline across observations, with the minimum and maximum values ranging from -268.57 to -126.54 lb/acre. A similar trend is observed for $\text{Revenue_change_Tmax2}$, with a mean loss of \$122.07 per acre, showing the economic impact of temperature rise, with losses varying from \$218.88 to \$60.36 per acre. In contrast, changes in precipitation (Yield_change_prcp) show a positive effect on yield, with an average increase of 76.75 lb/acre. The variability in this effect is evident from the standard deviation (48.63), and yield improvements range from 3.04 to 181.36 lb/acre. The corresponding revenue impact ($\text{Revenue_change_prcp}$) is also positive, with an average gain of \$51.01 per acre, ranging from \$1.80 to \$123.33 per acre. Consequently, these results highlight the contrasting effects of heat stress and precipitation on cotton

TABLE 3 | Summary of results by year: Extended yield and revenue data table (2005–2018).

Year	Yield change TmaxGS2 (lb/acre)	Revenue change TmaxGS2 (\$)	Yield change Prcp (lb/acre)	Revenue change Prcp (\$)
2005	−126.54	−60.36	74.42	35.50
2006	−160.29	−74.53	74.52	34.65
2007	−155.16	−92.01	3.04	1.80
2008	−191.28	−91.43	70.51	33.70
2009	−151.15	−95.07	71.97	45.27
2010	−268.57	−218.88	44.73	36.45
2011	−199.48	−176.14	110.44	96.46
2012	−245.76	−178.18	98.69	82.68
2013	−152.37	−118.70	63.84	31.15
2014	−195.57	−119.88	31.75	19.46
2015	−148.47	−90.86	118.15	72.31
2016	−159.10	−108.19	183.03	123.33
2017	−157.98	−108.37	104.01	83.93
2018	−250.32	−176.41	25.98	18.26

Note: Predicting yield by an additional unit of TmaxGS2 (1°C) and precipitation (mm/day).

TABLE 4 | Summary statistics for yield and revenue.

Variable	Mean	Standard deviation	Min	Max
Yield_change_Tmax2 (lb/acre)	−183.0447	44.02398	−268.5688	−126.5379
Revenue_change_Tmax2 (\$)	−122.0718	46.74484	−218.8836	−60.35856
Yield_change_prcp	76.74847	48.63457	3.038327	181.3643
Revenue_change_prcp (\$)	51.00788	36.49647	1.801728	123.3277

production, simultaneously emphasising the economic vulnerability to rising temperatures and the potential mitigating role of adequate precipitation.

Below is Table 5, and it provides a more specific interpretation of the yearly predictions using the standard error approach, focusing on the certainty or uncertainty of the predictions based on the yield changes for TmaxGS2 and Precipitation (Prcp).

Assessment of Yield Changes for TmaxGS2: The predictions for TmaxGS2 show consistently large negative yield changes across all years, with values ranging from −184.48 lb/acre in 2005 to −348.90 lb/acre in 2010. The substantial declines indicate a strong negative relationship between increasing temperatures and cotton yield. Given the magnitude of these changes, we can infer a high level of certainty regarding the detrimental impact of rising temperatures on yields.

Revenue Impact from TmaxGS2: Correspondingly, revenue changes reflect this trend, with predictions showing significant losses. The revenue changes range from −\$87.99 to −\$284.35, emphasising the economic ramifications of heat stress. The consistency of large negative revenue impacts suggests that these predictions are reliable, as they indicate a clear and negative effect of increasing TmaxGS2 on cotton profitability.

Evaluation of Precipitation Predictions: With regard to Precipitation, the yield changes present a more mixed picture. Positive yield changes occur in years like 2011 (135.41 lb/acre) and 2016 (194.93 lb/acre), while lower or negative yield changes appear in other years, such as 2007 (12.99 lb/acre) and 2018 (37.50 lb/acre). The variability in yield changes due to precipitation suggests greater uncertainty in these predictions. This variability could imply that while increased rainfall may benefit yields in some years, the benefits are not consistent.

Revenue Changes for Precipitation: The revenue changes for precipitation also reflect this inconsistency, with values ranging from \$7.70 to \$119.56. The fluctuating revenue changes imply that the economic implications of precipitation are less certain compared to the pronounced impacts of temperature increases. This indicates that while increased precipitation can lead to significant gains in certain years, the overall reliability of these predictions is lower due to the observed variability.

3.5 | Validation of Yield Loss Estimates Through Experimental Evidence

In support of our panel data findings, results from a recent meta-analysis of experimental studies, as shown in Table 6

TABLE 5 | Summary of results by year: Extended yield and revenue data table (2005–2018).

Year	Yield change TmaxGS2 (lb/acre)	Revenue change TmaxGS2 (\$)	Yield change Prcp (lb/acre)	Revenue change Prcp (\$)
2005	−184.48	−87.99	83.78175	39.96
2006	−212.93	−99.01	83.49	38.82
2007	−220.54	−130.78	12.99	7.703
2008	−258.16	−123.40	81.58	38.99
2009	−215.70	−135.67	82.39	51.82
2010	−348.90	−284.35	58.02	47.28
2011	−256.36	−226.36	135.41	119.56
2012	−323.89	−234.82	149.73	108.55
2013	−214.09	−166.78	53.65	41.79
2014	−273.48	−167.64	43.95	26.94
2015	−218.86	−133.94	129.72	79.39
2016	−239.70	−162.99	194.93	132.55
2017	−228.67	−156.87	84.31	57.84
2018	−320.23	−225.12	37.50	26.36

Note: Predicting yield by an additional unit of 1 standard deviation on TmaxGS2 and Precipitation.

TABLE 6 | Effect of heat stress on cotton yield parameters.

Variables	SMD	95% CI	SE	p value	Heterogeneity		
					Q	df	I ² /%
Boll number	−2.02	−2.66, −1.38	0.33	< 0.001	268.37	42	84
Boll retention	−1.10	−1.61, −0.58	0.26	< 0.001	197.14	51	74
Boll weight	−0.38	−0.83, 0.06	0.23	0.094	237.26	52	78
Seed yield	−0.20	−0.54, 0.14	0.17	0.251	74.15	42	43
Lint yield	−1.81	−3.37, −0.25	0.80	0.023	320.33	15	95

Abbreviations: CI, confidence interval; df, degree of freedom; I², inconsistency index; P, probability; Q, Cochran statistic; SE, standard error; SMD, standardised mean difference.

confirm the detrimental effects of heat stress on key cotton yield parameters. The meta-analysis, which synthesised results from multiple controlled experiments, revealed that heat stress significantly reduced boll number (standardised mean difference [SMD] = −2.28, $p < 0.001$), boll retention (SMD = −2.53, $p < 0.001$), and boll weight (SMD = −0.88, $p < 0.001$)—all are critical contributors to final yield. Most notably, lint yield, which directly reflects economic returns, showed a large negative effect (SMD = −3.37, $p < 0.001$), highlighting the magnitude of heat stress damage during key developmental stages. Below is Table 6, and it expresses the statistical analysis of the impact of heat stress on cotton yield.

While the SMD is a unitless measure and does not directly translate into percentage yield loss, a large negative effect size typically indicates a substantial real-world reduction in yield. Based on our findings and supporting literature, where yield reduction is estimated by multiplying the SMD by the standard deviation, we observed a yield loss of approximately 3.24 lbs/acre across growth stages under severe heat stress, depending on baseline growing conditions. This aligns with

our panel data analysis, which showed that a 1°C increase in maximum temperature during the flowering stage (TmaxGS2) was associated with an average 5.5% decline in yield and revenue losses of up to \$219 per acre in the Southern United States. The consistency between experimental and observational results enhances the validity and robustness of our conclusions and highlights the urgent need for heat-resilient cotton varieties, adaptive irrigation strategies, and economic risk mitigation tools such as crop insurance.

4 | Conclusion and Recommendation

This study highlights the profound impact of heat stress on cotton production in the Southern United States, revealing significant correlations between rising temperatures and declining yields. The analysis demonstrates that maximum temperatures during critical growth stages, particularly flowering and boll development, adversely affect cotton yields, with each degree of increase leading to substantial losses.

Furthermore, the economic ramifications of these yield reductions are considerable, as lower cotton production directly translates to diminished revenues for farmers. Given the importance of cotton as a vital cash crop, the findings accentuate the urgent need for adaptive strategies to mitigate these adverse effects.

To enhance resilience against heat stress, it is crucial for stakeholders—farmers, agricultural scientists, and policymakers—to collaborate on the development of heat-tolerant cotton varieties and innovative farming practices. Implementing efficient irrigation systems, optimising planting schedules, and employing agronomic techniques that improve soil health can help buffer the effects of rising temperatures. Additionally, increasing awareness and training for farmers on these adaptive measures can empower them to make informed decisions that safeguard both their yields and revenue.

Policymakers must also consider financial support mechanisms to aid cotton producers facing the economic challenges posed by heat stress. This may involve the creation of crop insurance programmes tailored to protect against yield losses due to extreme weather events and fluctuations in climate. Moreover, investing in research initiatives focused on understanding the interplay between heat stress, yield, and revenue can guide future agricultural policies and practices. By proactively addressing these challenges, the agricultural sector can better position itself to adapt to changing climate conditions, ensuring the sustainability and profitability of cotton production in the years to come.

5 | Limitations and Future Research Section

This study employs a multilevel panel data approach to assess the impact of heat stress on cotton yield and revenue across the Southern United States. While panel data provides valuable insights into long-term trends and regional variability, it has inherent limitations in controlling for all exogenous variables. One of the challenges of using panel data is the potential for omitted variable bias, as certain unobserved factors (e.g., soil health, farm management decisions, and genetic variability of cotton varieties) may influence yield outcomes. To mitigate these concerns, we apply a multilevel regression model that accounts for both spatial and temporal heterogeneity, incorporating fixed effects and robust standard errors to control for location-specific and time-variant unobserved factors. Additionally, the inclusion of weather variables (e.g., TmaxGS2, precipitation, and growing degree days) enhances the model's ability to capture climatic effects on yield. Despite these strengths, experimental field studies provide controlled insights into the physiological and biochemical responses of cotton to heat stress. To complement this study, we recently conducted a meta-analysis of experimental studies on the impact of heat stress on cotton yield and its parameters. The findings from this meta-analysis offer further validation of the econometric estimates presented in this study. Future research should focus on integrating experimental evidence with panel data models, allowing for more precise yield predictions and better adaptation strategies for cotton farmers. By combining large-scale empirical data with experimental validation, future studies can

enhance the precision of climate impact assessments and strengthen adaptation measures for cotton production in a warming climate.

Author Contributions

Oduniyi Oluwaseun Samuel: conceptualisation, methodology, writing the original draft, formal analysis, visualisation, data curator, review and editing. **McCallister Donna:** supervision, review, resources and project administration, investigation. **Gao Long:** methodology, formal analysis, review and editing. **Bastos Leonardo:** data curator, resources and project administration. **Jagadish S. V. Krishna:** supervision, review, resources and project administration, investigation.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data supporting the findings of this study were obtained from publicly accessible sources, including the United States Department of Agriculture (USDA) and meteorological records. However, some portions of the dataset used in the analysis are subject to data-sharing restrictions and may not be publicly distributed. Researchers interested in accessing the processed dataset may contact the corresponding author for further information.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.